EXP 11: Write a program for the task of Credit Score Classification

CODING:

```
# Credit Score Classification using KNN

import math

# Training Dataset
X = [
   [1, 25000],
   [2, 30000],
   [3, 45000],
   [4, 60000],
   [5, 80000],
   [6, 100000],
   [2, 35000],
   [5, 75000]
]

Y = [
   'Low', 'Low', 'Medium', 'Medium',
   'High', 'High', 'Low', 'High'
]

# Euclidean Distance
def distance(a, b):
   total = 0
```

```
    for i in range(len(a)):
        total += (a[i] - b[i]) ** 2

    return math.sqrt(total)

# KNN Classification
def predict(sample, k=3):
    distances = []

    for i in range(len(X)):
        d = distance(sample, X[i])
        distances.append((d, Y[i]))

    distances.sort(key=lambda x: x[0])

    neighbours = []

    for i in range(k):
        neighbours.append(distances[i][1])

    # Find Majority Class
    prediction = max(set(neighbours),
                     key=neighbours.count)

    return prediction

# Display Training Data
print("Credit Score Classification")
```

```
print("\nTraining Dataset:")

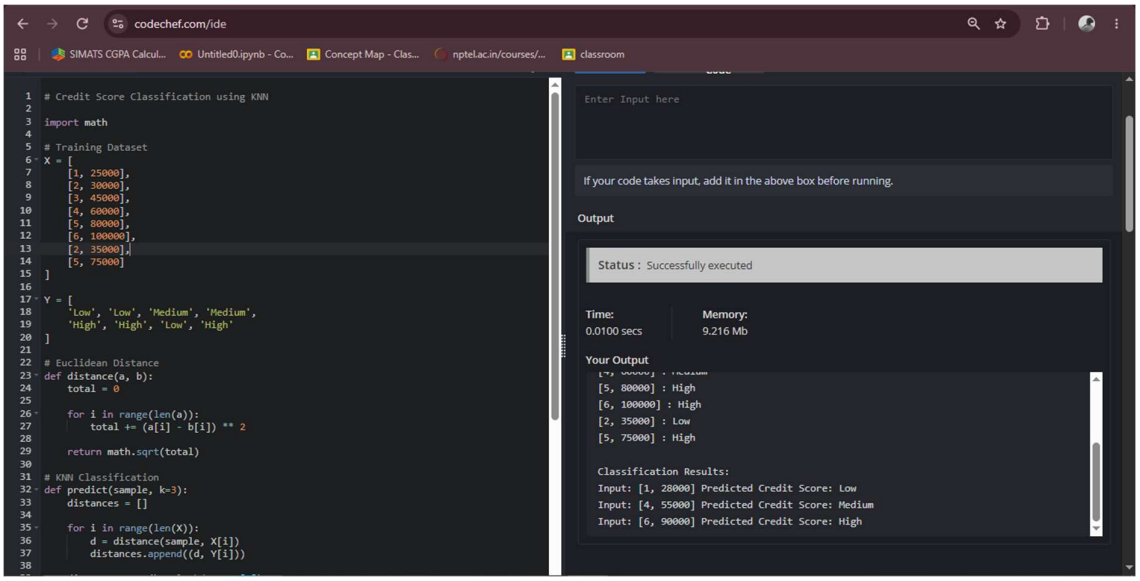
for i in range(len(X)):
    print(X[i], ":", Y[i])

# Test Data
test_data = [
    [1, 28000],
    [4, 55000],
    [6, 90000]
]

print("\nClassification Results:")

for sample in test_data:
    result = predict(sample)
    print("Input:", sample,
          "Predicted Credit Score:", result)
```

OUTPUT:


codechef.com/ide
SIMATS CGPA Calcul...
Untitled0.ipynb - Co...
Concept Map - Clas...
nptel.ac.in/courses/...
classroom
Credit Score Classification using KNN
import math
Training Dataset
X = [
[1, 25000],
[2, 30000],
[3, 45000],
[4, 60000],
[5, 80000],
[6, 100000],
[2, 35000],
[5, 75000]
]
Y = [
'Low', 'Low', 'Medium', 'Medium',
'High', 'High', 'Low', 'High'
]
Euclidean Distance
def distance(a, b):
total = 0
for i in range(len(a)):
total += (a[i] - b[i]) ** 2
return math.sqrt(total)
KNN Classification
def predict(sample, k=3):
distances = []
for i in range(len(X)):
d = distance(sample, X[i])
distances.append((d, Y[i]))
Enter Input here
If your code takes input, add it in the above box before running.
Output
Status : Successfully executed
Time:
0.0100 secs
Memory:
9.216 Mb
Your Output
[5, 80000] : High
[6, 100000] : High
[2, 35000] : Low
[5, 75000] : High
Classification Results:
Input: [1, 28000] Predicted Credit Score: Low
Input: [4, 55000] Predicted Credit Score: Medium
Input: [6, 90000] Predicted Credit Score: High